Research on Driving Factors of COVID-19 Protective Behaviours in Australia Based on Deep Learning

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Introduction

- Personal protective behaviours played a crucial role.
- Understanding main drivers is important for making public health policy.

Background

Multiple Factors Shape Behaviour

Protective behaviours are influenced by

- Demographic
- Psychological
- Policy factors
- Vaccination
- COVID-19 case numbers
- Other features

Drivers of mask mandates and protective behaviour changes in Australia

- Protective behaviours are not fixed
- The face mask mandates may influence the drivers of protective behaviours in Australia. (Ryan et al. 2025, 8–13)

MLP has shown potential in structured data

- MLP(Multi-Layer Perceptron)s can achieve competitive or even stronger performance on structured data.

Research Questions

- How do vaccination rates and COVID-19 case numbers influence the protective behaviours in Australia?
- How does the performance of deep learning models compare to traditional models in predicting protective behaviours?

Data-YouGov Behaviour Tracker

Imperial College London YouGov Behaviour Tracker dataset

- Demographic and psychological factors (such as gender and age)
- protective behaviours (such as mask wearing and social distancing)
- Comorbidities
- other features

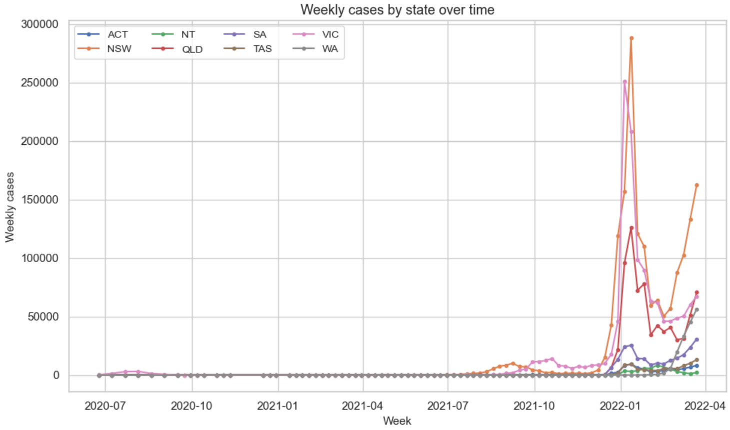
Data-Oxford COVID-19 Government Response Tracker

Oxford COVID-19 Government Response Tracker dataset

- The state level response policies of governments around the world aimed at preventing the spread of COVID-19 from March 2020 to January 2023

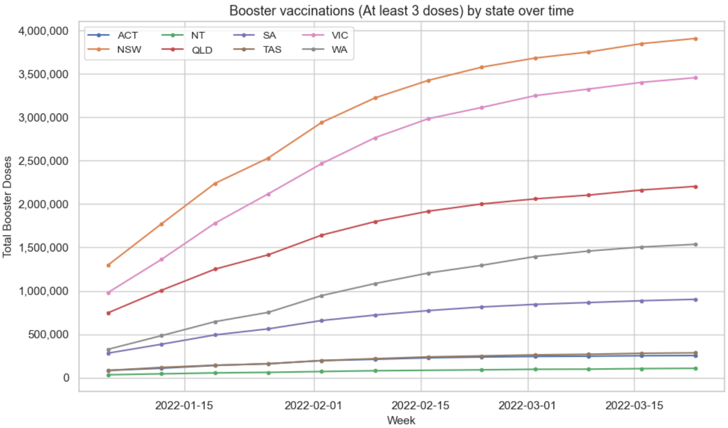
Data-Pandemic Case number

Consolidated dataset for COVID-19 case numbers, which compiles and verifies COVID-19 case data from Australian federal, state, and territory health departments. (covid19data.com.au 2023a)



Data-Vaccination Data

Vaccine dataset compiled by a third-party team to match the timing and states of the YouGov survey. (covid19data.com.au 2023b)



Preprocessing

Government Mandates

- Calculated the14-day rolling average of each state's index.
- Rolling average of a certain state reached 3 was considered entered mandate.

Removal of Features with High Missingness

- 10,781 as threshold (Ryan et al. 2025, 3)
- Over threshold were dropped from the dataset.

Preprocessing

Repairing Data Gaps Caused by Survey Changes

- YouGov modified the survey between February 10 and October 18, 2021.
- Located this time window using a date mask
- Filled in with independent 'N/A' categories.

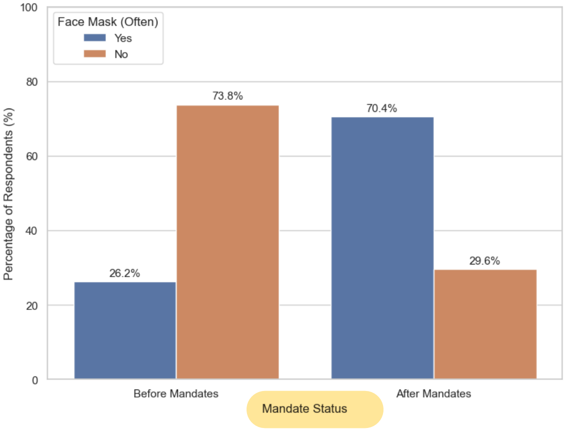
Predictive Variables

Calculate the two-week average for the following features and convert into binary:

- Mask wearing
- Protective behaviours
- Protective behaviours other than masks.

Methods-Overresampling

- This study introduced the random over sampler technique through `imblearn.pipeline`.



Methods-Cross Validation

- A single validation set is highly susceptible to the randomness of the initial data split.
- To prevent overfitting during model training, this study implemented a 5-fold cross validation.

Evaluation metrics and interpretation

Main metric

AUC-ROC serves as the primary metric for tuning and model comparison.

Further performance comparison

These metrics offer deeper insights beyond AUC-ROC

- Precision
- Recall
- F1 score

Interpretation

- SHAP summary plots are applied to the best-performing model

Methods-Models

Logistic Regression

- Regularization strength
- Regularization type

Classification Trees

- Maximum depth of the tree
- Minimum samples per leaf
- Minimum samples to split a node

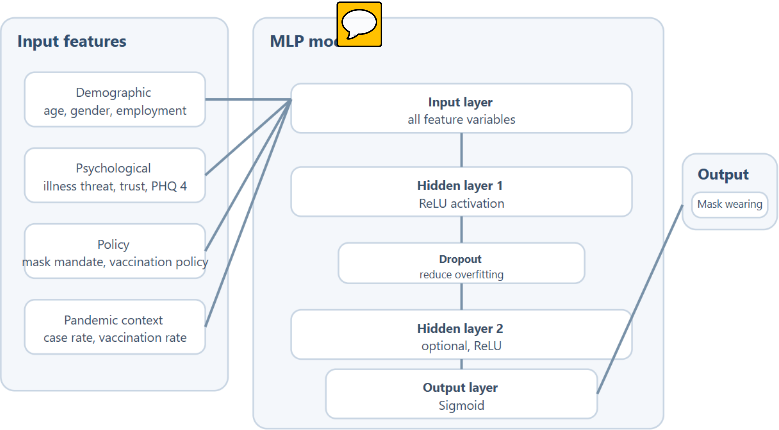
XGBoost

- Learning_rate
- Number of trees
- Maximum depth of each tree
- Number of features per split

Multilayer Perceptron

MODEL STRUCTURE

Proposed MLP architecture for predicting mask wearing



Training: 5 fold cross validation on the training set

Evaluation: F1 score and ROC AUC on the testing set

Interpretation: SHAP values for the two best performing models

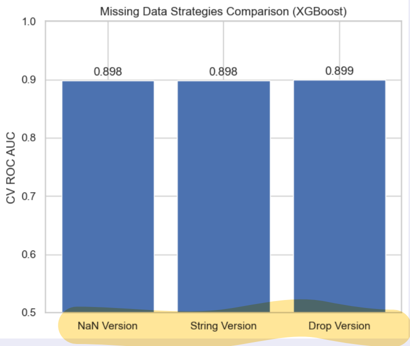
Results- Experiment on handling structural missing values

Content

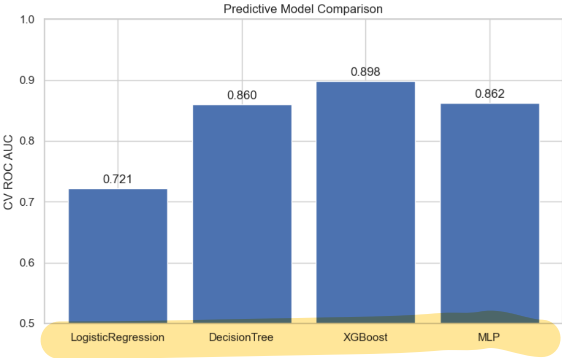
Compared three ways on handling structural missing values:

- Retaining np.nan
- Imputing an "N/A" category
- Dropping features directly

Results



Results- Model Comparison



- Logistic Regression performed the worst.
- MLP performed only slightly better than the Decision Tree.
- The best-performing model was XGBoost.

Future

- Conduct better model tuning or investigate feature importance further.
- Incorporate additional data features.
- Expand the scope of the research to include other countries.